



OFFICE OF POSTGRADUATE COURSES AND MASTER'S PROGRAMMES

It is hereby
certified that:

Dr HORNSBY-ODOI LEMUEL
born on 01/03/1996 in ACCRA (GHANA)

passed, on 09/03/2026, with a mark of 100/110, the Level II Master's examination in SUSTAINABLE SHIP AND SHIPPING 4.0, DEPARTMENT OF INDUSTRIAL ENGINEERING.

It is further certified that, in accordance with current regulations, the duration of the course of study is eighteen months.

To obtain the aforementioned qualification, he sat and passed the following examinations:

Course	Date Exam	Grade University	Credits	(*)
11986 FINAL EXAM	09/03/2026	Passed	3	016
26070 INTERNSHIP	15/02/2026	Completed	15	016
47566 DIGITAL TWIN IN MARINE SYSTEMS	06/11/2025	27	6	V89
47567 SIMULATION AND OPTIMISATION OF SHIPBUILDING PROCESSES	11 March 2025	26	6	V89
47485 INTELLIGENT DECISION SUPPORT SYSTEMS	04/07/2025	19	6	V89
47484 MANOEUVRABILITY AND SHALLOW WATER SHIP HYDRODYNAMICS	20 June 2025	26	3	V89
47489 AUTONOMOUS MARINE VEHICLES	16/06/2025	25	6	V89
47486 TRANSPORT ECONOMICS	12/06/2025	25	3	V89
47483 INTRODUCTION TO MARINE COMPUTATIONAL FLUID DYNAMICS (CFD)	06/06/2025	23	6	V89
47482 INDUSTRY 4.0 ENABLING TECHNOLOGIES	02/06/2025	27	6	V89
45488 STRUCTURAL ANALYSIS OF SHIPS AND OFFSHORE STRUCTURES (SHIP STRUCTURES)	24/02/2025	22	6	016
45485 SHIP DESIGN	04/02/2025	30	6	016
45489 VIRTUAL PROTOTYPING	30/01/2025	28	6	016
45484 REGULATORY FRAMEWORK FOR MARITIME INDUSTRY 4.0	16/01/2025	28	3	016
45490 ROBOTICS	20/12/2024	24	6	016
45486 ALTERNATIVE FUELS FOR SHIPPING	18/12/2024	26	3	016

(*) 016: UNIVERSITY OF NAPLES FEDERICO II (*) V89: UNIVERSITY OF A CORUÑA

(*) 016: UNIVERSITY OF NAPLES FEDERICO II

It is further certified that the title of the thesis discussed is as follows: "Hybrid deep learning and predictive model control for the safe autonomous mooring of yachts in confined waters" – Supervisor MARCOS MIGUEZ GONZALEZ.

Pursuant to Article 15 of Law 183/2011, this certificate may not be issued to the authorities public administration bodies or private operators of public services within the territory of the Italian Republic. The data contained in this certificate are extracted from the computerised database of the University Centre for Information Services (CSI).

The signature of the head of the office is omitted pursuant to Article 3 of Legislative Decree No. 39 of 12 February 1993.



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Serial No.: SH4000078

Naples, 13 March 2026
Office

The Head of the

ENG. FLAVIO CICCARELLI

This certificate is issued on plain paper for purposes for which the law does not require a revenue stamp
(Presidential Decree No. 642 of 26 October 1972, Annex B and subsequent amendments).